

GE261004	DESIGN THINKING	L	T	P	C
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COURSE OBJECTIVES:					
To introduce students to the principles of design thinking and enable them to identify real-world problems, generate innovative ideas, and develop user-centered solutions through creativity, collaboration, and iterative design processes.					
UNIT I	Introduction to Design Thinking				6
Design thinking concept and importance – Engineering design vs traditional problem solving – Innovation and creativity – Design thinking process overview – Human-centered design – Role of engineers in societal problem solving – Case studies of successful innovations.					
UNIT II	Empathy and Problem Definition				6
Understanding users and stakeholders – Empathy mapping – Observation techniques – Interviews and need finding – Problem identification – Framing problem statements – Point of View (POV) statements – Problem reframing.					
UNIT III	Ideation and Creative Thinking				6
Idea generation techniques – Brainstorming – Mind mapping – SCAMPER technique – Lateral thinking – Concept selection methods – Feasibility and innovation evaluation – Team ideation exercises.					
UNIT IV	Prototyping and Testing				6
Concept visualization – Low-fidelity prototyping – Sketching and mock-ups – Rapid prototyping concepts – Testing methods – User feedback – Iterative design process – Design improvement strategies.					
UNIT V	Innovation, Entrepreneurship and Sustainable Design				6
Introduction to innovation and entrepreneurship – Value proposition – Sustainable and inclusive design – Social innovation – Basics of product development cycle – Pitch preparation and storytelling – Ethical considerations in design.					
TOTAL: 30 PERIODS					
COURSE OUTCOMES:					
At the end of the course, students will be able to					
C01 :	Explain the principles and stages of the design thinking process.				
C02 :	Identify user needs and formulate problem statements using empathy-based approaches.				
C03 :	Generate and evaluate innovative ideas using structured ideation techniques.				
C04 :	Develop basic prototypes and test solutions through iterative design methods.				
C05 :	Present sustainable and innovative engineering solutions addressing real-world problems.				
TEXTBOOKS :					
1	Tim Brown, Change by Design: How Design Thinking Transforms Organizations and Inspires Innovation, Harper Collins, 2012.				
2	Idris Mootee, Design Thinking for Strategic Innovation: What They Can't Teach You at Business or Design School, 1st Edition, John Wiley & Sons, 2013.				

REFERENCES:	
1	Jeanne Liedtka & Tim Ogilvie, Designing for Growth: A Design Thinking Toolkit for Managers, 1st Edition, Columbia Business School Publishing, 2011.
2	Nigel Cross, Engineering Design Methods: Strategies for Product Design, 4th Edition, John Wiley & Sons, 2008.
3	Donella H. Meadows, (2015), "Thinking in Systems -A Primer", Sustainability Institute.
	https://www.ideo.com/pages/design-thinking#process

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PS01	PS02	PS03
C01	2	1	–	–	–	1	–	–	–	–	2			
C02	1	3	2	–	–	2	–	2	1	–	2			
C03	1	2	3	–	1	1	–	2	1	–	2			
C04	1	2	3	2	2	1	–	2	1	–	2			
C05	2	2	3	1	1	3	2	2	3	2	3			